

EXPERIMENT- 06

Python Code:

```
06.py > ...
1  def mod_inverse(a, m):
2      for x in range(1, m):
3          if (a * x) % m == 1:
4              return x
5      return -1
6
7  p1, p2 = 4, 19
8  c1, c2 = 1, 20
9
10 diff_p = (p1 - p2) % 26
11 diff_c = (c1 - c2) % 26
12
13 inv_diff_p = mod_inverse(diff_p, 26)
14
15 a = (diff_c * inv_diff_p) % 26
16 b = (c1 - a * p1) % 26
17
18 print(f"Calculated Keys: a = {a}, b = {b}")
19
20 a_inv = mod_inverse(a, 26)
21
22 ciphertext = input("Enter ciphertext: ").upper()
23 decrypted_text = ""
24
25 for char in ciphertext:
26     if char.isalpha():
27         c = ord(char) - 65
28         p = (a_inv * (c - b)) % 26
29         decrypted_text += chr(p + 65)
30     else:
31         decrypted_text += char
32
33 print("Decrypted Text:", decrypted_text)
```

Output:

```
PS D:\College\Crypto\Experiments> python -u "d:\College\Crypto\Experiments\06.py"
Calculated Keys: a = 3, b = 15
Enter ciphertext: KBWWF
Decrypted Text: HELLO
PS D:\College\Crypto\Experiments> 
```